

THEOREM. *In an integral domain $\mathbf{A}$, for every polynomial P in the variable x with coefficients in $\mathbf{A}$, $P \in \mathbf{A}[x]$, and every $r \in \mathbf{A}$, there exist unique $\mathcal{Q} \in \mathbf{A}[x]$ and $\mathcal{R} \in \mathbf{A}$ such that*

$$P(x) = (x - r) \mathcal{Q}(x) + \mathcal{R},$$

where

$$\mathcal{R} = P(r).$$

○ 1. The existence

Let $P(x)$ be a polynomial in $\mathbf{A}[x]$ of degree n ,

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = \sum_{i=0}^n a_i x^i,$$

where each coefficient $a_i \in \mathbf{A}$. Evaluating it at $x = r$ gives

$$P(r) = a_n r^n + a_{n-1} r^{n-1} + \cdots + a_1 r + a_0 = \sum_{i=0}^n a_i r^i.$$

Consider the difference

$$P(x) - P(r) = \sum_{i=0}^n a_i x^i - \sum_{i=0}^n a_i r^i.$$

By the linearity of summation, the two expressions can be combined term-by-term. Moreover, the constant terms for $i=0$ cancel out ($x^0 = r^0 = 1 \Rightarrow a_0 x^0 - a_0 r^0 = a_0 - a_0 = 0$). The difference $P(x) - P(r)$ becomes

$$\begin{aligned} \sum_{i=0}^n a_i x^i - \sum_{i=0}^n a_i r^i &= (a_0 - a_0) + \sum_{i=1}^n (a_i x^i - a_i r^i) \\ &= \sum_{i=1}^n a_i (x^i - r^i). \end{aligned}$$

Each term $x^i - r^i$ in the above sum is a difference of two i -th powers. The following lemma on the factorization of $x^n - c^n$ for any $n \geq 1$ shows that every such difference can be factored by $(x - r)$. Applying the lemma term-by-term exposes a common factor $(x - r)$ across all the terms, so it can be factored out of the entire expression.

LEMMA. For any integer $n \geq 1$, the difference of two n -th powers, $x^n - c^n$, can be factored by $(x - c)$. That is, there exists a polynomial $S_{n-1}(x, c)$ such that

$$x^n - c^n = (x - c)S_{n-1}(x, c).$$

○ (by mathematical induction on n)

The **base case** $n = 1$ holds since

$$x^1 - c^1 = (x - c)(1),$$

where the constant $1 = S_0(x, c)$ is a polynomial as well.

Inductive Hypothesis. Assume for some arbitrary positive integer $k \geq 1$ there exists a polynomial $S_{k-1}(x, c)$ such that

$$x^k - c^k = (x - c)S_{k-1}(x, c).$$

Induction Step. To show that the lemma holds for $k + 1$ (assuming it holds for k), consider

$$x^{k+1} - c^{k+1},$$

that can be rewritten by adding $0 = -xc^k + c^kx$ and grouping as

$$x^{k+1} - xc^k + c^kx - c^{k+1} = (x^{k+1} - xc^k) + (c^kx - c^{k+1}).$$

Factoring out the common components from each group gives

$$x(x^k - c^k) + c^k(x - c).$$

Applying the inductive hypothesis to substitute $(x - c)S_{k-1}(x, c)$ for $(x^k - c^k)$ yields

$$x(x - c)S_{k-1}(x, c) + c^k(x - c).$$

Factoring out the common multiplier $(x - c)$ results in

$$(x - c)(xS_{k-1}(x, c) + c^k),$$

where $xS_{k-1}(x, c) + c^k = S_k(x, c)$ is a polynomial — since $S_{k-1}(x, c)$ is a polynomial by the inductive hypothesis, and polynomial rings are closed under multiplication and addition.

This completes the induction step, proving the lemma for all $n \geq 1$. ●

Explicitly, the $S(x, c)$ polynomials are

$$\begin{aligned}
1 &= S_0 \\
x(1) + c^1 &= x + c &= S_1 \\
x(x + c) + c^2 &= x^2 + xc + c^2 &= S_2 \\
x(x^2 + xc + c^2) + c^3 &= x^3 + x^2c + xc^2 + c^3 &= S_3 \\
x(x^3 + x^2c + xc^2 + c^3) + c^4 &= x^4 + x^3c + x^2c^2 + xc^3 + c^4 &= S_4 \\
&\dots \\
x(x^{n-2} + x^{n-3}c + \dots + xc^{n-3} + c^{n-2}) + c^{n-1} &= x^{n-1} + x^{n-2}c + \dots + xc^{n-2} + c^{n-1} = S_{n-1} \\
x(x^{n-1} + x^{n-2}c + \dots + xc^{n-2} + c^{n-1}) + c^n &= x^n + x^{n-1}c + \dots + xc^{n-1} + c^n = S_n
\end{aligned}$$

In more compact notation with the summation symbol

$$S_n(x, c) = x^n + x^{n-1}c + \dots + xc^{n-1} + c^n = \sum_{i=0}^n x^{n-i}c^i$$

and

$$xS_{n-1}(x, c) + c^n = x \sum_{i=0}^{n-1} x^{n-1-i}c^i + c^n = \sum_{i=0}^{n-1} x^{n-i}c^i + c^n = \sum_{i=0}^n x^{n-i}c^i = S_n(x, c).$$

The exact factorization is $x^n - c^n = (x - c) \sum_{i=0}^{n-1} x^{n-1-i}c^i$.

Several interesting properties characterize $S_n(x, c) = \sum_{i=0}^n x^{n-i}c^i$ as a *complete homogeneous symmetric polynomial* in two variables :

- ✓ it is *symmetric*, since $S_n(x, c) = \sum_{i=0}^n x^{n-i}c^i = \sum_{i=0}^n c^{n-i}x^i = S_n(c, x)$,
- ✓ it is *homogeneous* of degree n , because the degree of each monomial $x^{n-i}c^i$ is exactly $n-i+i=n$ for all i ,
- ✓ it is *complete* in the sense that it contains every possible n -th degree monomial in the variables x and c exactly once, each with a coefficient of 1.

The symmetry of $S_n(x, c)$ is already hinted at in the induction step. The proof inserts the zero $0 = -xc^k + c^kx$, but one may equally insert the zero $0 = -x^kc + cx^k$. Then $x^{k+1} - x^kc + cx^k - c^{k+1} = x^k(x - c) + c(x^k - c^k)$ and the resulting recurrence will be

$$S_k(x, c) = x^k + cS_{k-1}(x, c).$$

The recurrences $x^k + cS_{k-1}(x, c)$ and $xS_{k-1}(x, c) + c^k$ are dual under the swap $x \leftrightarrow c$.

Returning to $P(x) - P(r) = \sum_{i=1}^n a_i(x^i - r^i),$

the lemma yields*

$$x^i - r^i = (x - r)S_{i-1}(x, r)$$

for every $i \geq 1$.

* From a deeper algebraic viewpoint, $S_{i-1}(x, c)$ from the lemma belongs to the polynomial ring $A[x, c]$. Replacing the indeterminate c by the element $r \in A$ is the evaluation homomorphism $A[x, c] \rightarrow A[x]$. Thus $S_{i-1}(x, r)$ is a polynomial in $A[x]$.

Substituting these factorizations term-by-term gives

$$P(x) - P(r) = \sum_{i=1}^n a_i (x - r) S_{i-1}(x, r) = (x - r) \sum_{i=1}^n a_i S_{i-1}(x, r).$$

Since each $S_{i-1}(x, r)$ is a polynomial in $A[x]$ and polynomial sums are polynomials, the sum $\sum_{i=1}^n a_i S_{i-1}(x, r)$ belongs to the same polynomial ring $A[x]$. Define

$$\sum_{i=1}^n a_i S_{i-1}(x, r) = Q(x) \in A[x].$$

Hence
$$P(x) - P(r) = (x - r) Q(x)$$

or equivalently
$$P(x) = (x - r) Q(x) + P(r)$$

— the latter exhibits the intended representation with $\mathcal{R} = P(r) \in A$. Ergo, such $Q \in A[x]$ and $\mathcal{R} \in A$ exist. ●

It remains to show that $Q(x)$ and $\mathcal{R}$ are unique.

○ 2. The uniqueness

Assume that there exist polynomials $Q', Q'' \in A[x]$ and constants $R', R'' \in A$ such that

$$P(x) = (x - r) Q'(x) + R' \quad \text{and} \quad P(x) = (x - r) Q''(x) + R''.$$

Subtracting the second representation from the first gives

$$(x - r) Q'(x) + R' - (x - r) Q''(x) - R'' = 0,$$

which simplifies to

$$(x - r)(Q'(x) - Q''(x)) = R'' - R'.$$

Evaluating this polynomial identity at $x = r$, the factor $(x - r)$ becomes $r - r = 0$, so the left-hand side vanishes and reveals that the constant difference is zero :

$$0 = R'' - R',$$

that is

$$R' = R''.$$

Returning to the polynomial identity before evaluation and putting $R' = R''$ into it yields

$$(x - r)(Q'(x) - Q''(x)) = 0.$$

Because A is an integral domain, the polynomial ring $A[x]$ is an integral domain as well and therefore has no zero divisors. Moreover, $x - r$ is a nonzero polynomial in $A[x]$. Since its product with $Q'(x) - Q''(x)$ is the zero polynomial, the absence of zero divisors in $A[x]$ implies that

$$Q'(x) - Q''(x) = 0,$$

giving

$$Q'(x) = Q''(x).$$

Thus
$$Q' = Q'' \quad \text{and} \quad R' = R'',$$

showing that the intended representation

$$P(x) = (x - r)Q(x) + \mathcal{R}$$

is unique in the choice of Q and $\mathcal{R}$. ●